8) | Lecture 15-16 | (2-4) Tree |

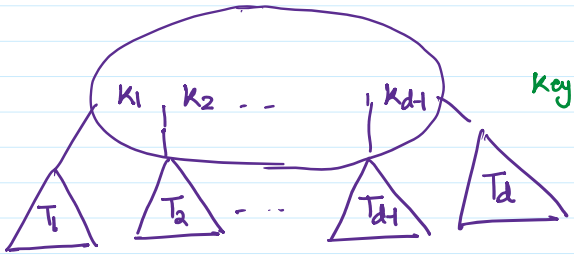
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Multiway Search Tree (Key, Val)

Each internal node in multiway search tree T

- has at least two children
- Stores a collection of item of form (key, val) — key, data — children
- node

- if an internal node contains (d-1) item, then it should have d children



$Key(T_1) \le K_1 \le Key(T_2) \cdots \le K_{d-1} \le Key(T_d)$

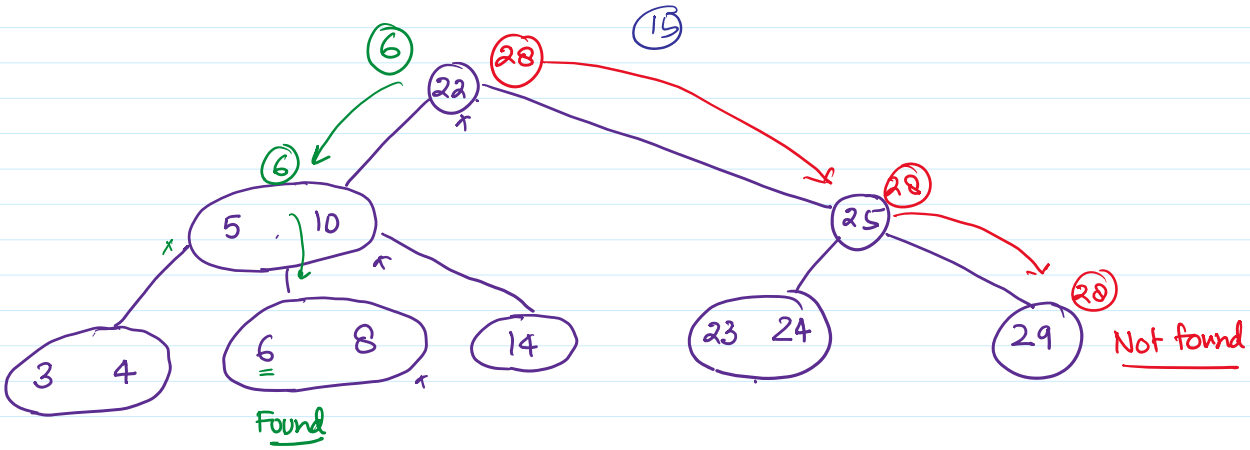
$K_1 \le K_2 \cdots \le K_{d-1}$

for subtree T_i $K_{i-1} \le Key(T_i) \le K_i$

(2-4) tree (d=4)

- Each internal node should have at least 2 children
- if an internal node contains (d-1) item, then it should have d children

- Each internal node can have at most 3 keys and at most 4 children
- All leafs are at same level



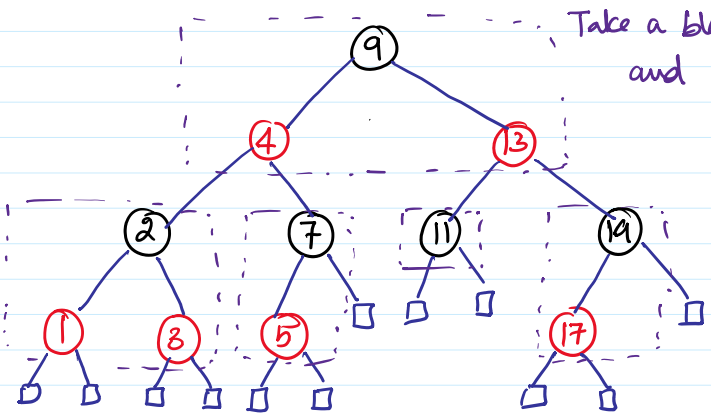
Search 28

Let n be number of node in (2-4) tree. Let h be the height of (2-4) tree

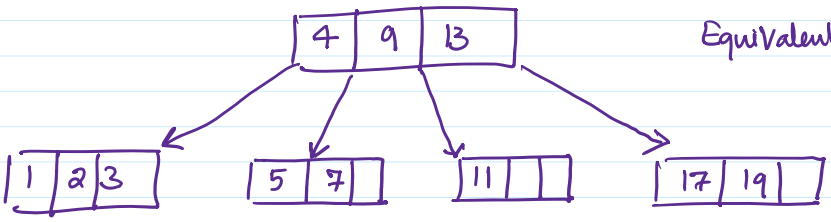
$$\log_4 n \leq h \leq \log_2 n$$

$$h = \Theta(\log n)$$

Both Red Black and (2-4) tree are equivalent



Take a black node and its red children and make one node of (2-4) tree



Equivalent (2-4) tree